

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a65b835b112753fca.jsonl`  
- SHA-256: `b3a0685e27b09f3735c3c0fb51bcffa0cbde8404b0df8e3347e41b7a4b4fc8e4`  
- Source modified: `2026-05-31T04:19:04+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:18:17.933Z)

Source Extractor

Research question: "Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.
- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON

transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly."

Fetch and extract key claims from this source:

****URL:**** <https://github.com/OpenGVLab/InternVL/issues/900>

****Title:**** InternVL2.5-4B license (MIT repo, Qwen-Research component) ? GitHub issue

****Found via:**** Model-weights license minefield (commercial-safety, decisive) search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-05-31T04:18:19.888Z)

I'll fetch the GitHub issue page and extract the relevant claims.

3. user (2026-05-31T04:18:29.613Z)

InternVL2.5-4B License Discussion - Issue #900

****Issue Details:****

- ****Title:**** About the license of internVL2.5-4B model
- ****Date:**** February 12, 2025
- ****Author:**** NaokiAsatani

What the Issue Presents

The user raises a licensing discrepancy regarding the InternVL2.5-4B model weights on Hugging Face. According to the original issue body:

The Hugging Face page states: "This project uses the pre-trained Qwen2.5-3B-Instruct as a component, which is licensed under the Apache License 2.0."

However, the user clarifies that "the license for Qwen2.5-3B-Instruct is not Apache 2.0, but rather the Qwen RESEARCH LICENSE."

Key Licensing Distinction

The user emphasizes an important distinction: the Qwen RESEARCH LICENSE "prohibits commercial use," and asks for clarification on whether InternVL2.5-4B is eligible for commercial use given this component dependency.

Status

****No maintainer or contributor responses are visible**** in the provided content. The issue shows "Closed" status, but the actual discussion thread and any official clarifications are not included in this excerpt.

The MIT license applies to the repository code itself, but the model weights incorporate a Qwen component with potentially more restrictive terms.

4. assistant (2026-05-31T04:18:33.294Z)

The fetch succeeded but the WebFetch summarizer noted the maintainer responses may not have been captured. Let me try the GitHub API to get the full comment thread, since this is a public issue.

5. user (2026-05-31T04:18:35.381Z)

```
{"body":"Thank you very much for providing such an excellent model under the MIT license. \n\nI would like to ask one question regarding the licensing. \n\nIn the Hugging Face page for the InternVL2_5-4B model, the license section includes the following statement: \n\n\"This project is released under the MIT License. This project uses the pre-trained Qwen2.5-3B-Instruct as a component, which is licensed under the Apache License 2.0.\" \n\n\nHowever, the license for Qwen2.5-3B-Instruct is not Apache 2.0, but rather the Qwen RESEARCH LICENSE. Since the Qwen RESEARCH LICENSE prohibits commercial use, I would like to confirm whether the InternVL2_5-4B model is eligible for commercial use under these circumstances. \n\n\nThank you for your clarification.\n\n\", \"comments\":0, \"created_at\":\"2025-02-12T04:05:09Z\", \"state\":\"closed\", \"title\":\"About the license of internVL2.5-4B model\", \"user\":\"NaokiAsatani\"}
```

6. assistant (2026-05-31T04:18:38.136Z)

The GitHub API confirms the issue body exactly and shows `comments: 0` ? there are no maintainer responses; the issue was closed without any reply. This is itself an important finding: the licensing question was raised but never officially answered by the InternVL maintainers.

Let me verify there are truly no comments via the comments endpoint.

7. user (2026-05-31T04:18:41.532Z)

0

8. assistant (2026-05-31T04:18:47.664Z)

Confirmed: zero comments. The issue was raised on 2025-02-12 and closed with no maintainer reply.

This is a primary source ? the original issue text on the official OpenGVLab/InternVL GitHub repository, retrieved directly from the GitHub API. The body text is verbatim from the issue author. I now have enough to extract the falsifiable claims.

Key findings relevant to muneem's research question (commercial-license safety of InternVL2.5 weights):

1. The InternVL HF model card claims Qwen2.5-3B-Instruct is "Apache License 2.0" ? but the issue author asserts this is wrong; it's the Qwen RESEARCH LICENSE.
2. The Qwen Research License prohibits commercial use, casting doubt on commercial eligibility of InternVL2.5-4B.
3. The issue received zero responses and was closed ? meaning the license uncertainty was never

resolved by the maintainers (important for "call out license uncertainty explicitly").

9. user (2026-05-31T04:18:54.054Z)

Structured output provided successfully

10. assistant (2026-05-31T04:19:04.811Z)

Done. I fetched GitHub issue #900 (verified via the GitHub API, which confirmed the verbatim issue body and `comments: 0`, `state: closed`).

Summary of what I found, relevant to muneem's commercial-license-safety question:

Source quality: **primary** ? the original issue text from the official OpenGVLab/InternVL repository, retrieved directly from the GitHub API (not a summarizer's paraphrase).

Key takeaways:

- The InternVL2.5-4B HF model card claims its Qwen2.5-3B-Instruct component is Apache-2.0, but the issue asserts this is **incorrect** ? the real license is the **Qwen Research License**, which **prohibits commercial use**.
- This means the MIT label on the InternVL repo does **not** clear the **weights** for commercial use; the smaller InternVL2.5 sizes built on Qwen2.5-3B inherit a non-commercial restriction. This directly supports the research-question warning to verify weights licenses, not just repo licenses, and flags InternVL2.5-4B as **license-unsafe for commercial use**.
- Notably, the maintainers **never responded** ? the issue was closed with zero comments, so the discrepancy remains officially unresolved by the project. That is a concrete "license uncertainty" data point to call out explicitly.

Caveat: this is a single user's assertion in an unanswered issue. The underlying fact (Qwen Research License on Qwen2.5 3B/14B/72B sizes; only 0.5B/1.5B/7B/32B are Apache-2.0) should be cross-checked against Qwen's own license files ? but the issue is solid evidence that the InternVL2.5-4B card's Apache-2.0 attribution is contested.